

Name: _____

Date: _____

PERCENTAGES, (COMPOUND INTEREST AND DEPRECIATION)

LO: I can calculate compound interest and depreciation over multiple years.

Compound interest and depreciation

Compound interest means interest is added to the total each year, so the next year's interest is calculated on a **larger amount**. The value grows by a multiplier each year.

Depreciation is the opposite: the value **decreases** by a percentage each year. Use a multiplier less than 1.

Multipliers

●	5% increase × 1.05	add the percentage as a decimal
●	3% decrease × 0.97	subtract from 1
●	£1000 at 4% for 2 years = 1000×1.04^2	use power for years
●	5% increase × 0.05	that gives 5% OF, not increase

Key idea

Final amount = Original $\times$ (multiplier)ⁿ where n is the number of years.

$$A = P \times (1 \pm r)^n$$

P = principal, r = rate as decimal, n = years

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - compound interest

£2000 is invested at 5% compound interest per year. Find the value after 3 years.

$$\text{Multiplier} = 1.05$$

$$\text{Value} = 2000 \times 1.05^3$$

$$= 2000 \times 1.157625 = \text{£}2315.25$$

£2315.25

Raise the multiplier to the power of the number of years.

Example 2 - depreciation

A car worth £12,000 depreciates by 15% each year. Find its value after 2 years.

$$\text{Multiplier} = 1 - 0.15 = 0.85$$

$$\text{Value} = 12000 \times 0.85^2$$

$$= 12000 \times 0.7225 = \text{£}8670$$

£8670

For depreciation, subtract the rate from 1 to get the multiplier.

Example 3 - finding years

£5000 is invested at 10% compound interest. After how many years will it exceed £7000?

$$\text{Year 1: } 5000 \times 1.1 = 5500$$

$$\text{Year 2: } 5500 \times 1.1 = 6050$$

$$\text{Year 3: } 6050 \times 1.1 = 6655; \text{ Year 4:}$$

$$6655 \times 1.1 = 7320.50$$

After 4 years

Keep multiplying until the target is exceeded.

Example 4 - comparing simple and compound

Compare the interest on £3000 at 4% for 3 years: simple vs compound.

$$\text{Simple: } 3000 \times 0.04 \times 3 = \text{£}360$$

$$\text{Compound: } 3000 \times 1.04^3 = 3000 \times$$

$$1.124864 = \text{£}3374.59$$

Compound interest = £374.59 vs simple £360

Compound gives £14.59 more

Compound interest always gives more because interest earns interest.

YOUR TURN – PRACTICE (deliberate practice)

1. Compound interest basics

Find the final amount for each investment.

- a) £1000 at 3% for 2 years
- b) £5000 at 6% for 3 years
- c) £2500 at 4% for 2 years
- d) £800 at 10% for 3 years

2. Depreciation

Find the value after depreciation.

- a) £10,000 car, 20% per year, after 2 years
- b) £500 phone, 30% per year, after 2 years
- c) £18,000 car, 12% per year, after 3 years
- d) £3000 laptop, 25% per year, after 2 years

3. Finding the multiplier

Write the multiplier for each situation.

- a) 8% increase
- b) 15% decrease
- c) 2.5% increase
- d) 40% decrease

4. How many years?

Work out after how many years each condition is met.

- a) £1000 at 10%: when does it exceed £1300?
- b) £2000 at 20%: when does it exceed £3000?
- c) £8000 depreciates 10%: when is it below £6000?
- d) £5000 at 5%: when does it exceed £6000?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Compound interest uses a multiplier raised to the power of years ☐

Correct answer: _____

Reason: _____

b) A 20% depreciation multiplier is 0.20 ☐

Correct answer: _____

Reason: _____

c) Compound interest always gives more than simple interest ☐

Correct answer: _____

Reason: _____

d) After 2 years at 10%, you have gained exactly 20% ☐

Correct answer: _____

Reason: _____

e) Depreciation means the value decreases each year ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A savings account offers **3.5% compound interest**. You invest **£4500**. How much will you have after **5 years**? How much interest have you earned in total?

Expression: _____

Simplified: _____

b) A new car costs **£22,000** and depreciates by **18% per year**. After how many complete years will the car first be worth less than **£10,000**?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

